

What this file is, in five minutes

The aim. This is the **plan** for the next experiment — actually running the Berkeley dwarfs through your memory-capture — together with four experts' **prediction** of what that run will find. It is not results. It answers one thing: *before we spend server time, what should we capture, what can we realistically learn, and how will one cheap test tell us whether the big campaign is worth it?*

THE WHOLE STORY IN A BREATH

The snapshot fires every 500 ms. A fast compute kernel sweeps its whole array *thousands* of times in that half-second, and the snapshot blends every sweep into one “everything got touched” picture. So the 13 dwarfs you wanted to tell apart blur down to only about **6–8 distinguishable behaviors**. The guaranteed result is a map of which dwarfs are twins; the only open question is whether a few can be *un-blurred*; and one afternoon of testing decides it.

YOUR THREE QUESTIONS, ANSWERED

1. Is he saying we get another deliverable — the confusion matrix?

Yes. Two outcomes, and *both* hand you a real result:

- If the dwarfs separate cleanly → you get a working **classifier**. The win.
- If they blur together → the **confusion matrix is the deliverable**. It is a table: “I ran dwarf X; the memory-signal called it Y.” It shows exactly which dwarfs memory-writes *can* and *cannot* tell apart. That is a real, publishable finding — not a failure. It is the floor you cannot fall below: you never walk away empty-handed.

2. Is that what we see if we run the dwarf?

Yes, exactly. Run a dwarf, look at its write-signal, and the matrix shows what it looks *identical* to. It is the empirical picture of “what does each dwarf’s memory footprint actually look like, and which ones are indistinguishable twins.” It turns the whole question from a guess into a measured chart.

3. Are they saying 500 ms is too big an interval?

Yes — that is the central physical finding, for the compute kernels. In 500 ms a fast kernel sweeps its array thousands of times; the snapshot ORs all those sweeps into one flat frame, so the internal *motion* disappears. The camera blinks too slowly to catch the moving parts.

But here is the twist that matters: **the fix is not “shrink the interval.”** 500 ms is the fixed production cadence, and a finer one means far more overhead and data. So instead of speeding up the camera, they **slow down the subject** — make the kernel *bigger* so a single sweep takes longer than 500 ms.

Like photographing a spinning fan: you can't make the shutter faster, so you slow the fan until a single blade holds still long enough to see.

That trick only works where the compute grows faster than the memory it uses — QR and LU qualify (grow them $\sim 2\times$ – $4\times$ and one sweep finally spans several snapshots). Stencils and their kind never can: to slow them you'd need tens of gigabytes, far past the 1 GiB guest. Those stay a permanent blur — you will only ever see their *size*, never their *motion*.

THE PLAN ITSELF — ONE CHEAP AFTERNOON, GATED

1. **Gate 0** — pure arithmetic (done): which kernels can be grown enough to un-blur? QR and LU yes; stencil, DP, FFT no.
2. **Gate 1** — seconds on the server, no capture: run enlarged QR and LU, check they really slow down as predicted.

3. **Gate 2** — the one decisive check: capture `gemm` for ~15 s and see whether an array's changed pages sit together or scatter. Every “shape” feature depends on this.

4. **Gate 3** — the pilot proper: capture the enlarged QR/LU (do they show motion?), a few controls, and one deliberately-un-rescuable stencil as a trap that *should* stay blurred.

GO — enlarged QR/LU show their motion → the campaign is worth it, run the full set.

NO-GO — they stay blurred → motion is unrecoverable, and Part 2's honest result is the confusion matrix plus a size-and-magnitude classifier. Ship that, stop there.

Either way you have a thesis result. The pilot only decides whether it is the ambitious version (you can read a dwarf's motion) or the honest modest one (you can read its shape and name its twins).

One-page plain summary of the four-expert dwarf-pilot design. Fuller versions beside it:

`DWARF_PILOT_DESIGN_RAJA.{html,pdf}` (RAJA, full) and `DWARF_PILOT_DESIGN.{html,pdf}` (technical, with the tables and verbatim verdicts). Design only — no capture has run yet.